Paul Parisot

Software Engineer - Core C/C++ & Application Security

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About

Low-level systems engineer (C/C++, Assembly) specializing in performance-critical applications and memory safety.
Strong background in detecting and mitigating software vulnerabilities (buffer overflows, race conditions, memory leaks).
Experienced in Agile/TDD environments, with a practical understanding of relational databases and scripting for automation.

Skills

Core Languages: C, C++, Assembly, Python, Shell Unix, SQL
Data & Systems: Relational Databases (Query optimization, Joins), Linux Environments
Security & Testing: TDD, Vulnerability Remediation, Pentesting (HackTheBox, Nmap, Privilege Escalation)
Workflow: Agile/Scrum, Jira, Git, CI/CD, DevOps (Docker)

Professional Experience

Gno.land - Core Software Engineer (Open-Source System)

- Diagnosed and patched **critical memory-management vulnerabilities** (preventing panic/segfault scenarios).
- Enforced strict memory safety protocols and optimized concurrent processes to eliminate race conditions.
- Integrated extensive test suites under strict TDD and CI/CD policies, contributing **10k lines of code**.

Faurecia / Forvia - Software Engineer (Embedded C++ / Agile)

- Developed robust C++ network modules for embedded communication, ensuring secure low-level data exchange.
- Engineered a real-time navigation system showcased at **CES 2024**, delivering production-ready features ahead of schedule.
- Operated within an **Agile environment**, managing tasks via **Jira** and implementing rigorous code quality standards.

Epitech / IONIS - Technical Instructor (Systems & C/C++)

- Taught system-level programming and low-level architecture (C, C++, Asm) to over **400 students**.
- Evaluated code quality, focusing on memory optimization, structural precision, and the avoidance of runtime vulnerabilities.
- Produced clear technical documentation and structured assessment materials for complex programming concepts.

Technical Projects & Security

Sledge – Cloud Provider Migration System

Developed a generic migration tool involving complex data structure handling and database interactions.
Applied strong relational algebra principles to optimize **SQL queries** and manage cross-platform data integrity.

Skills: SQL Optimization, Relational Databases, Data Architecture, Scripting

Applied Security & Vulnerability Research

Participant in CTF challenges (HackTheBox, TryHackMe) focusing on boot2root and privilege escalation.
Analyzed binary decompilation and low-level software defects, gaining a deep comprehension of system architectures, Assembly, and memory layouts.
Practical experience with vulnerability detection tools (Nmap, Gobuster, Metasploit, ...) and network security (Port forwarding, Netcat, ...).

Skills: Pentesting, Reverse Engineering, Threat Analysis, Network Protocols

rodox – Modular Systems Engine

Designed a low-level engine architecture in C/C++ natively for **Linux**.
Implemented highly optimized custom memory allocators to prevent fragmentation and ensure runtime stability.

Skills: C++, Linux, Custom Memory Management, Pointer Arithmetic, Low-level Optimization

Education

Epitech, Paris 2022 – 2027
Master's Degree in Information Technology (Level-7 RNCP Certification)
Specialization: Systems Engineering and Software Architecture

McGill University, Montréal 2025 – 2026
Certificate in Management
Core coursework: Business Administration, Organizational Management